

Jason Mahaffey

Macon, GA · (316) 253-1281 · Jason.Mahaffey@gmail.com

LinkedIn: <https://www.linkedin.com/in/Jason-mahaffey-1bb7a977/>

SUMMARY

Data Scientist and Senior Program Leader with 15+ years of experience transforming complex analytics into clear, actionable insights for senior decision-makers. Adept at client-facing communication, executive presentations, and cross-functional collaboration. Expert in predictive modeling, statistical analysis, and data visualization with hands-on proficiency in Python, R, SQL, Spark, and Hadoop. Independently drives analytical initiatives from exploration to implementation. Proven ability to partner with technical and non-technical stakeholders to solve operational, strategic, and data-driven challenges.

CORE COMPETENCIES

- Client presentations and executive communication
 - Data science: machine learning, predictive modeling, statistical analysis
 - Data analytics & visualization: automated dashboards and reporting
 - Cross-functional collaboration across engineering, operations, and leadership teams
 - Program and project leadership, stakeholder alignment [Project Management Professional (PMI, Est. Apr 26)]
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PROFESSIONAL EXPERIENCE

Director, Enterprise Analytics & Portfolio Optimization

Air Force Reserve Command – Robins AFB, GA (2022–Present)

- Conducted statistical analysis across financial, workforce, asset availability, and utilization datasets (>100K records annually) and developed optimization models to identify cost–performance tradeoffs; presented findings to C-suite executives, improving budget efficiency by 12–18% and driving \$40–60M in annual cost avoidance across a \$480M portfolio.
- Analyzed asset performance, maintenance throughput, and lifecycle cost data and built predictive models linking investment decisions to operational outcomes;

informed executive tradeoffs that increased asset availability by 6–10% without increasing total spend.

- Integrated multi-source enterprise data into automated KPI dashboards and forecasting tools; improved year-end forecast accuracy from ~85% to >95% and reduced executive decision cycle time by ~30%.

Senior Program Analytics Manager, Operations & Modernization / Instructor Pilot

73rd Airlift Squadron – Scott AFB, IL (2016–2022)

- Conducted statistical assessments across reliability, maintenance, scheduling, and risk datasets and developed investment prioritization models; delivered recommendations that enabled on-time execution of a \$56M modernization initiative while reducing unplanned downtime by 15–20%.
- Analyzed operational performance, scheduling variance, and human-in-the-loop risk data to model execution reliability; briefed senior stakeholders, achieving >99% on-time completion rate across 80+ high-visibility global operations with zero critical failures.
- Unified logistics, staffing, and execution data into a single planning framework; applied data-driven scheduling logic that reduced coordination conflicts by ~25% and increased asset utilization by 10–15%.

Director, Analytics & Systems Validation

53d Test Management Group – Barksdale AFB, LA (2015–2016)

- Conducted statistical analysis across execution timelines, performance metrics, and cost datasets and developed end-to-end analytics frameworks; advised executive decision-makers on \$800M in capital investments, reducing time-to-decision and delivery timelines by ~20%.
- Partnered with engineering teams to analyze large-scale performance datasets (GB-scale) using statistical validation and anomaly detection models; reduced post-deployment defect rates by 25–30%.
- Built predictive risk and schedule models using historical execution data; improved outcome predictability by 15–20%, enabling earlier intervention and reduced downstream rework.

Head of Engineering Analytics & Reliability

49th Test & Evaluation Squadron – Barksdale AFB, LA (2012–2015)

- Conducted statistical assessments across high-reliability system performance and failure datasets and developed quantitative confidence and reliability models;

presented insights to senior executives, reducing analytical uncertainty by ~30% in high-stakes decisions.

- Applied time-series analysis and trend detection to long-horizon operational datasets; identified early degradation signals that improved risk detection lead time by ~20% compared to legacy methods.
 - Synthesized technical data, operational risk metrics, and policy constraints into executive-ready analytics products; reduced follow-on testing and rework requirements by ~15% through clearer, data-backed decisions.
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TECHNICAL SKILLS

Data Science & Analytics: Machine Learning, Predictive Modeling, Statistical Analysis, Time-Series Forecasting, Monte Carlo Simulation

Tools & Languages: Python, R, SQL, Spark, Hadoop, MATLAB, Jupyter, WatsonX.ai, Anaconda, RStudio [IBM Data Science Professional (Coursera, Est. Feb 26)]

Visualization: Dashboard design, automated reporting, executive briefings

Mathematical Expertise: Numerical Analysis, Optimization, Dynamical Systems

Cloud: Microsoft Azure (DP-100 certification, Ext. Feb 26)

Leadership: Air Force Senior Leader and Executive Leader Training Courses

EDUCATION

M.S., Mathematics — Texas A&M University (GPA 3.92)

M.A.S., Aeronautics — Embry-Riddle Aeronautical University (GPA 3.64)

B.S., Aeronautical Engineering (Minor: Math) — Air Force Academy (GPA 2.93)

AWARDS & CLEARANCE

- Southwest Illinois Leadership Council Innovation Airman of the Year (2020)
- 53d Wing Field Grade Officer of the Quarter (2015)
- Secret Clearance (Active Until 2028)